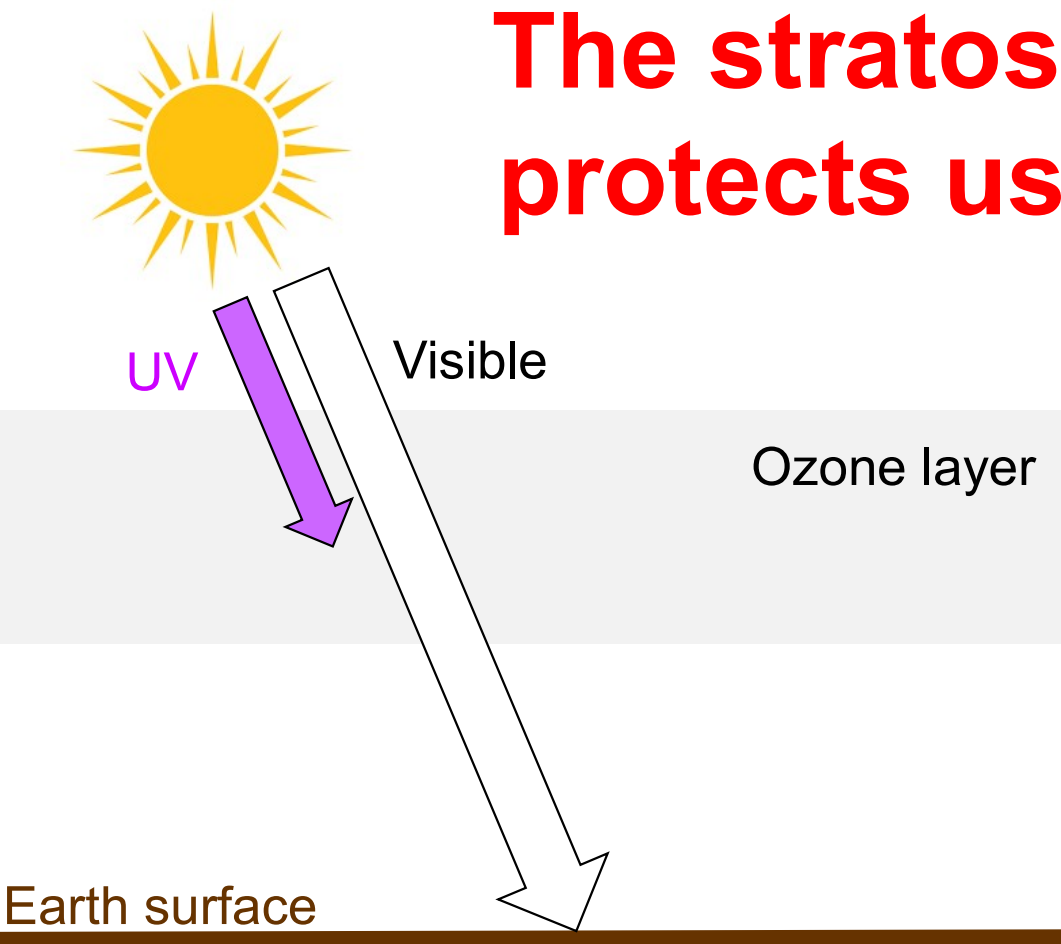


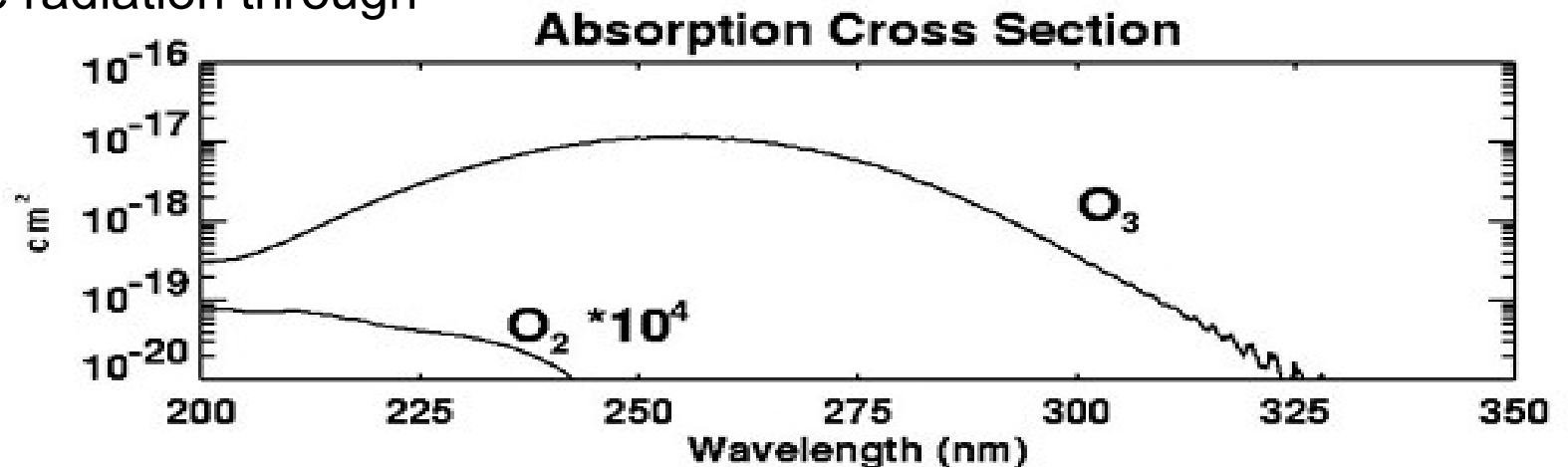
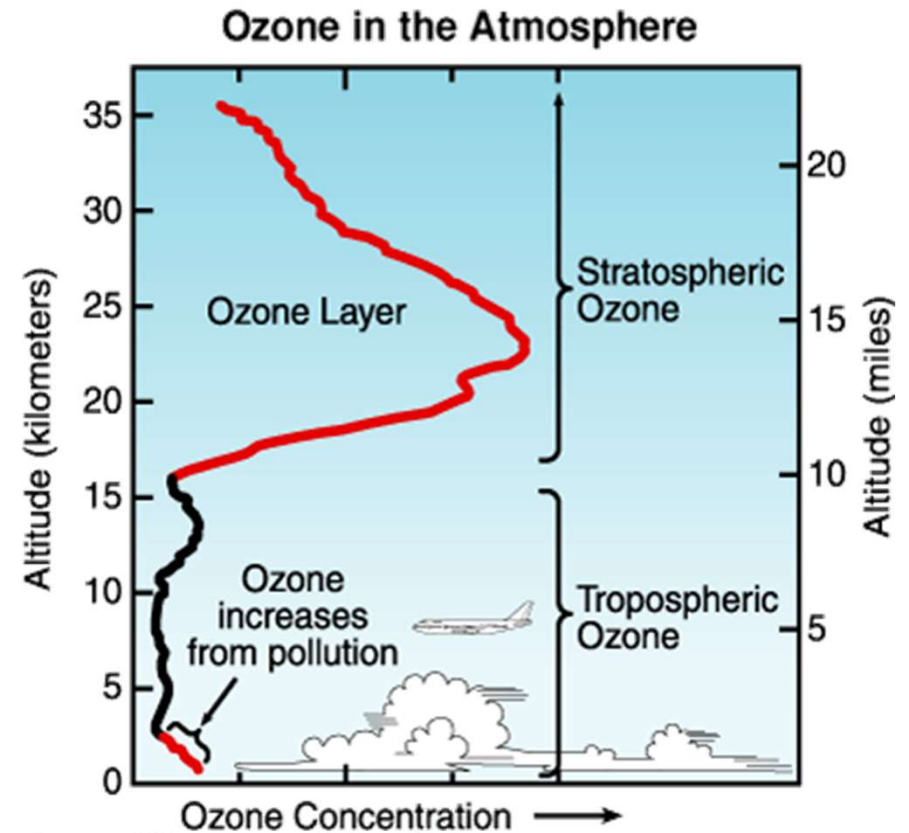
EES 301: Global biogeochemical cycles and Climate change



The stratospheric ozone layer protects us from UV radiation

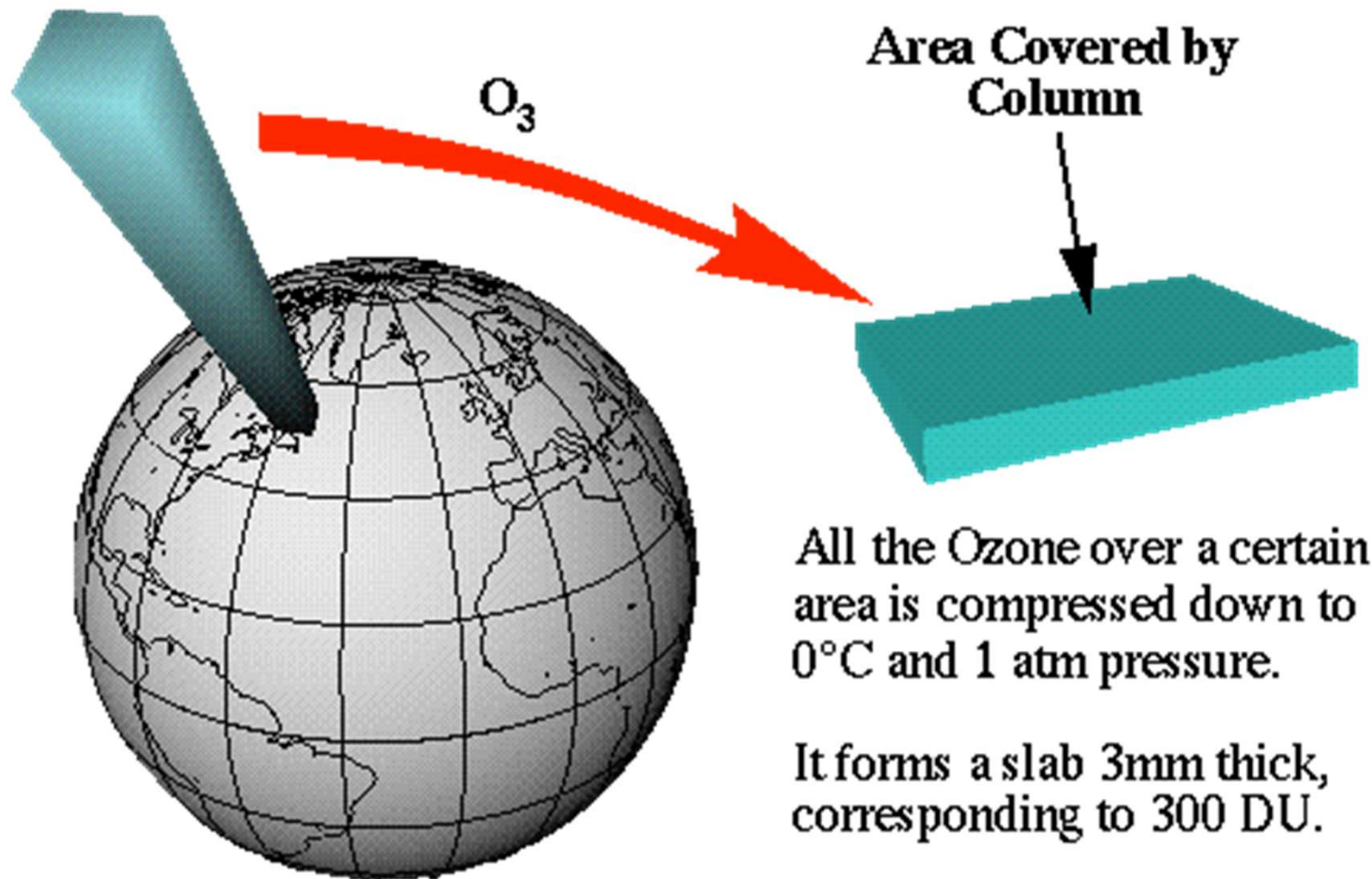


Ozone absorbs UV radiation while letting visible radiation through

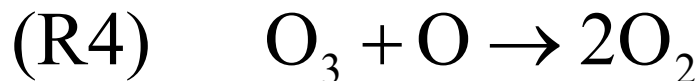
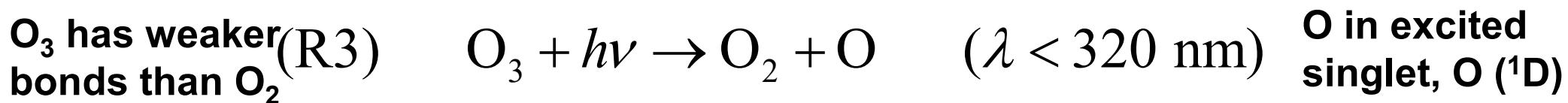
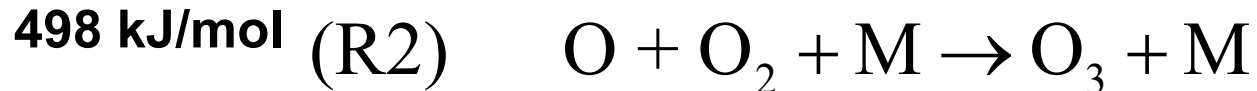
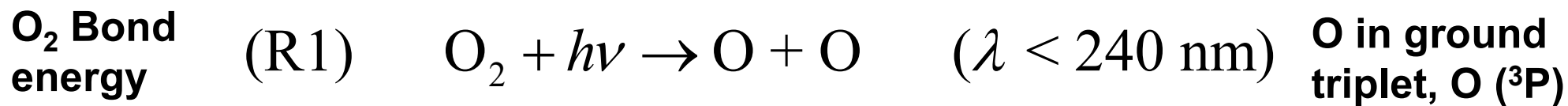


Dobson's observations (1930)

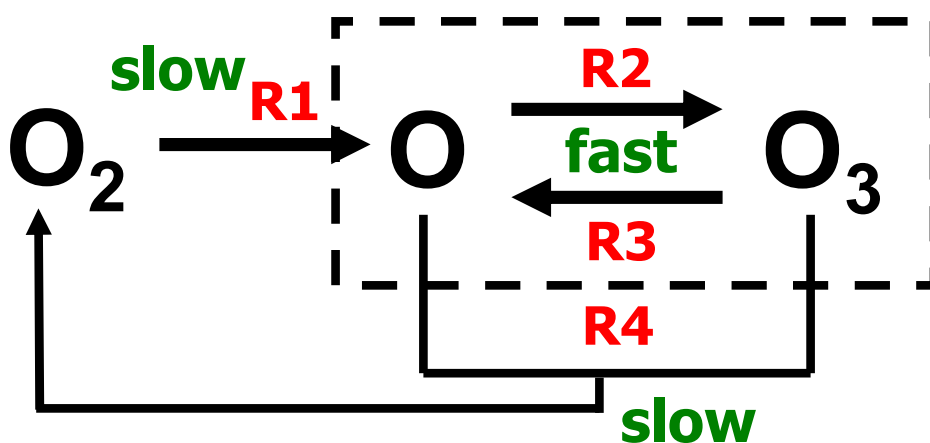
Ozone columns reported in units of 0.01 mm pure ozone at 1 atm and 0°C (STP), now known as Dobson unit (DU):
 $1 \text{ DU} = 2.69 \times 10^{16} \text{ molecules cm}^{-2}$



CHAPMAN MECHANISM FOR STRATOSPHERIC OZONE FORMATION



RATE CONSTANTS FOR R1-R4 HAVE BEEN MEASURED IN LABORATORY & FOUND TO BE:



Odd oxygen *family*
 $[\text{O}_x] = [\text{O}_3] + [\text{O}]$
 WHICH IS
 CONSUMED BY R4
 AND PRODUCED
 BY R1

SCHEMATIC OF CHAPMAN MECHANISM

STEADY-STATE ANALYSIS OF CHAPMAN MECHANISM

What is the Lifetime of O atoms in Chapman mechanism ?

$$\tau_O = \frac{[O]}{k_2[O][O_2][M] + k_4[O_3][O]} \approx \frac{1}{k_2 C_{O_2} n_a^2} \ll 1 \text{ s}$$

Short enough to assume chemical steady state for O (loss = prod) as Pressure, T, Radiation & ozone abundance won't change much in $t \sim 1\text{s}$: n_a is the number density of air

$$R2 = R3 \Rightarrow k_2[O][O_2][M] = k_3[O_3] \Rightarrow \frac{[O]}{[O_3]} = \frac{k_3}{k_2 C_{O_2} n_a^2} \ll 1$$

$\Rightarrow [O_x] \approx [O_3]$...so O_3 makes up most of the O_x budget

Lifetime of O_x : From R4; 2 coz 2 O_x used

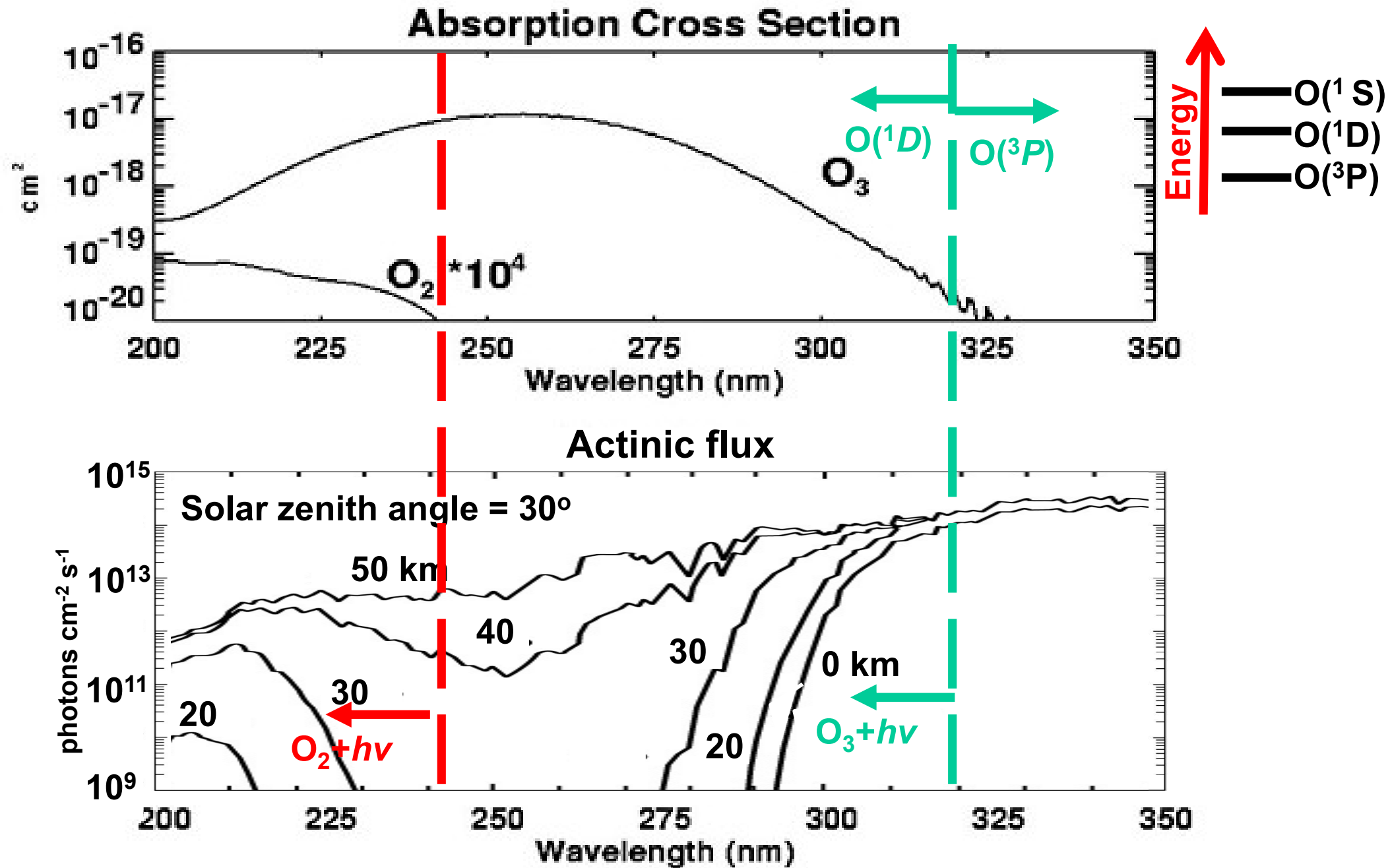
$$\tau_{O_x} = \frac{[O_x]}{2k_4[O_3][O]} \approx \frac{1}{2k_4[O]}$$

Assuming Steady state for O_x :

$$\begin{aligned} \frac{d}{dt}[O_x] &= \frac{d}{dt}([O_3] + [O]) = \frac{d}{dt}[O_3] + \frac{d}{dt}[O] \\ &= [R2 - R3 - R4] + [2R1 + R3 - R2 - R4] = 2R1 - 2R4 = 0 \\ 2R1 &= 2R4 \Rightarrow k_1[O_2] = k_4[O_3][O] \Rightarrow [O_3] = \left[\frac{k_1 k_2}{k_3 k_4} \right]^{\frac{1}{2}} C_{O_2} n_a^{\frac{3}{2}} \end{aligned}$$

Source: Daniel Jacob

Solar spectrum and absorption cross-sections



PHOTOLYSIS RATE CONSTANTS: VERTICAL DEPENDENCE

$$X + h\nu \rightarrow \dots \quad k = \int_0^\infty q_X(\lambda) \sigma_X(\lambda) I_\lambda d\lambda$$

quantum yield absorption X-section photon flux

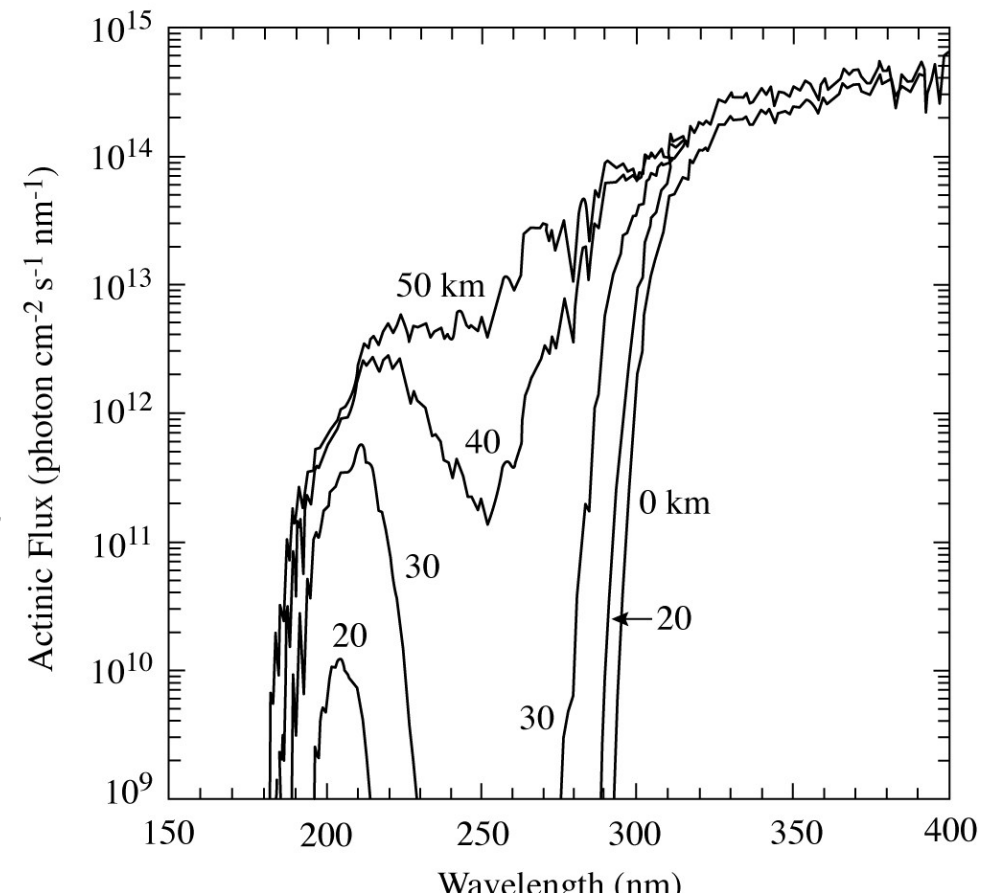
$I(z + dz)$

Values of I_λ are less at lower altitudes, so k_1 & k_3 decrease correspondingly

$I(z)$

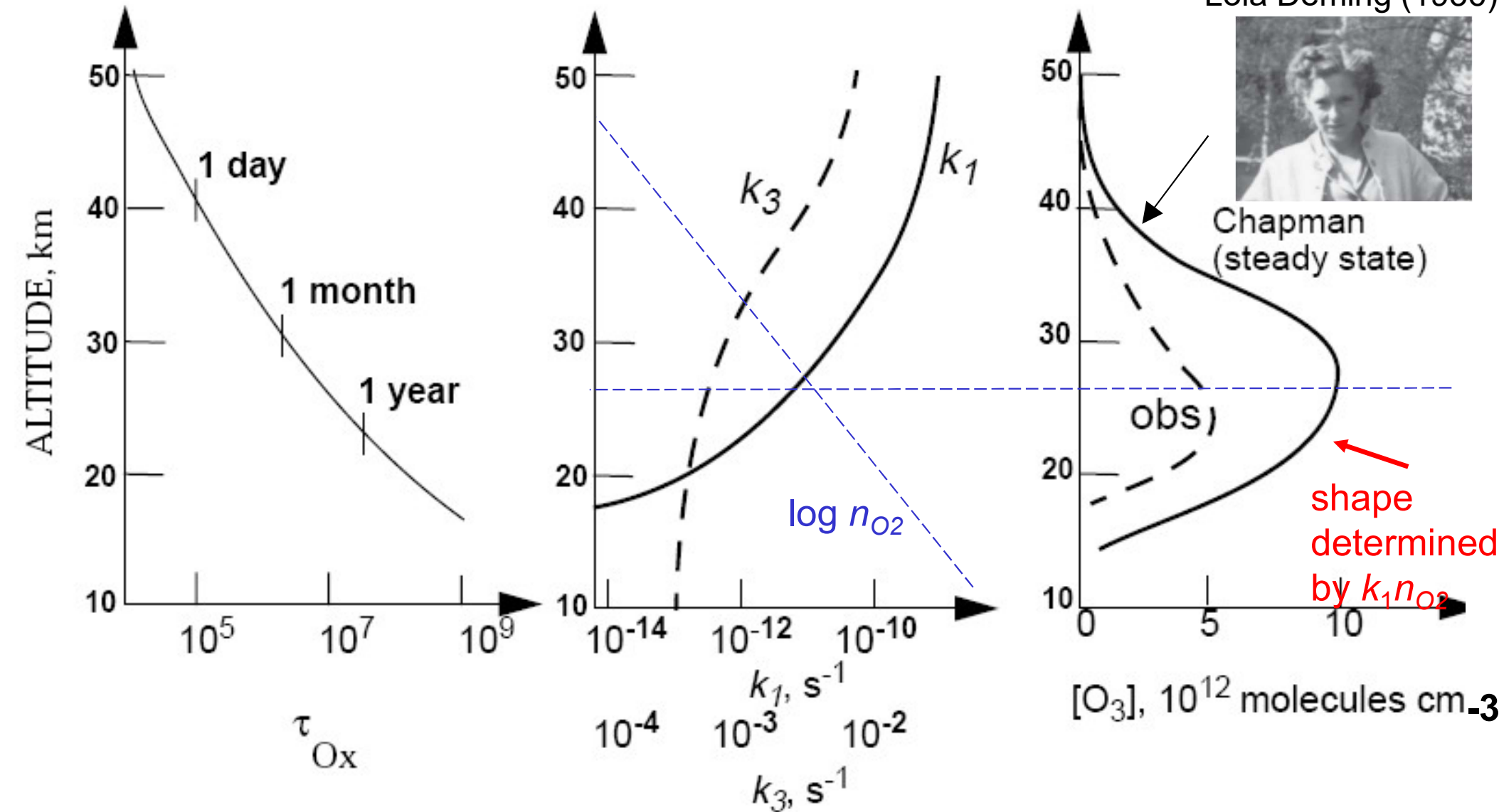
$$I(z) = I(\infty) e^{-\delta}$$

$$\delta = \int_z^\infty (\sigma_{O_2} n_{O_2}(z') + \sigma_{O_3} n_{O_3}(z')) dz'$$



Strong attenuation of photolysis rate constants k_1 and k_3 by ozone overhead;
steady-state $[O_3]$ must be computed numerically starting from top of atmosphere

Lola Deming (1936)

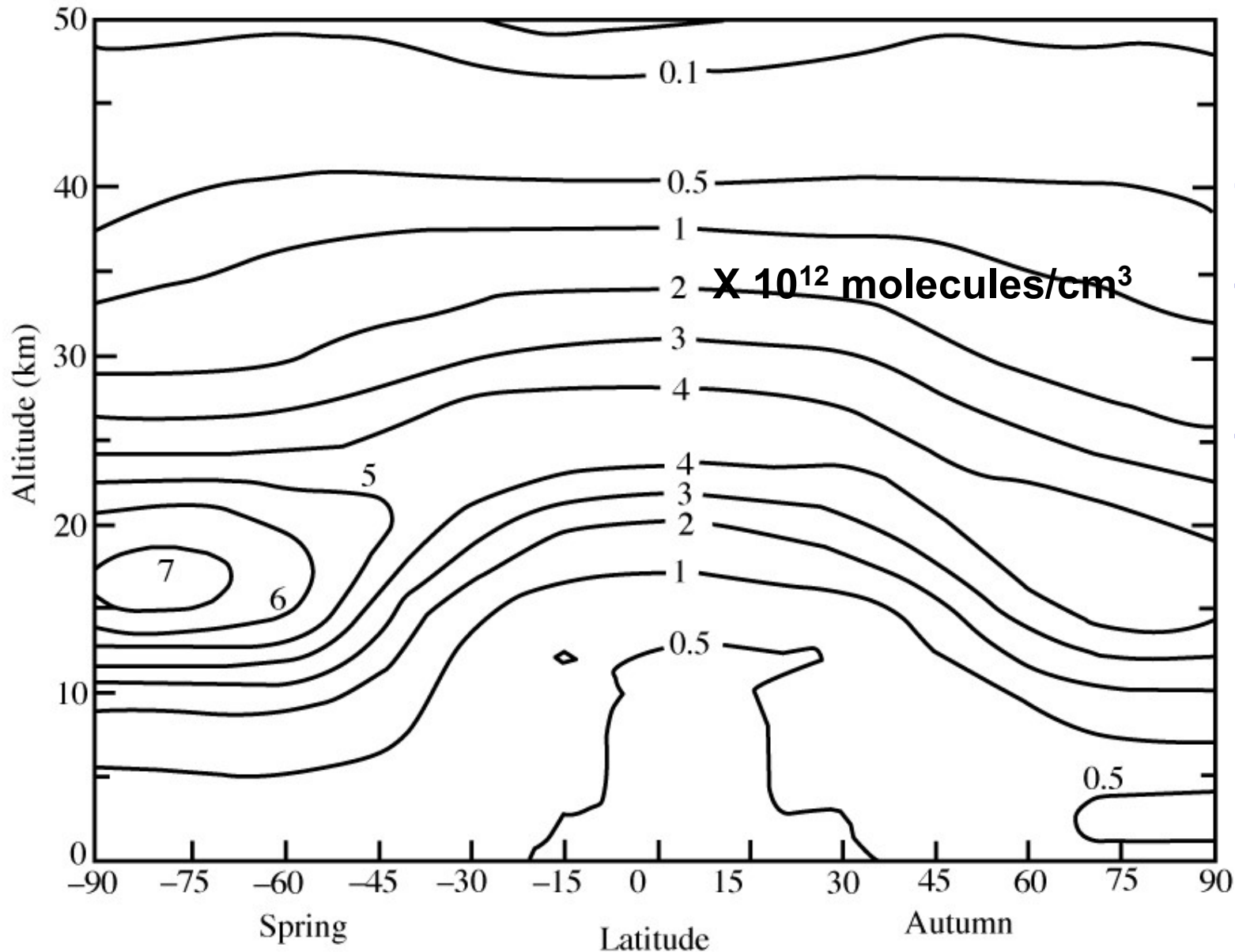


$$\tau_O = \frac{[O]}{k_2[O][O_2][M] + k_4[O_3][O]} \approx \frac{1}{k_2 C_{O_2} n_a^2} \approx 1 \text{ s}$$

Chapman mechanism reproduces shape, but is too high by factor 2-3 $\Rightarrow$ missing sink!

THE NATURAL OZONE LAYER

Based on ozonesonde observations in the 1970s



Observations confirmed the shape predicted by the Chapman mechanism but found it over predicted O₃ levels by more than a factor of 2



RADICAL REACTION CHAINS IN THE ATMOSPHERE

Initiation: non-radical $\longrightarrow$ radical + radical photolysis
thermolysis
oxidation by O(¹D)

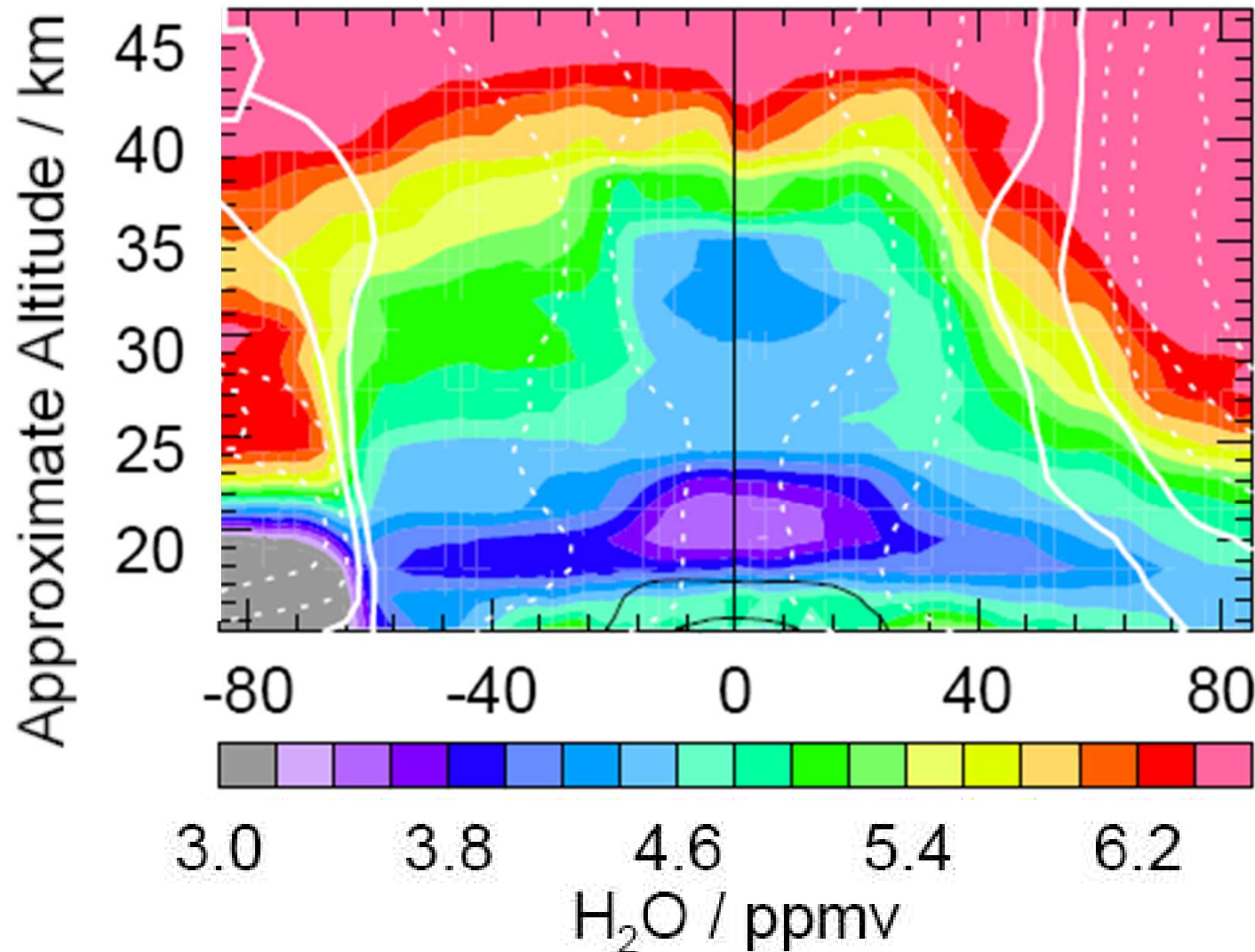
Propagation: radical + non-radical $\longrightarrow$ non-radical + radical bimolecular
redox reactions

Termination: radical + radical $\longrightarrow$ non-radical + non-radical radical redox
reaction

radical + radical + M $\longrightarrow$ non-radical + M 3-body recombination

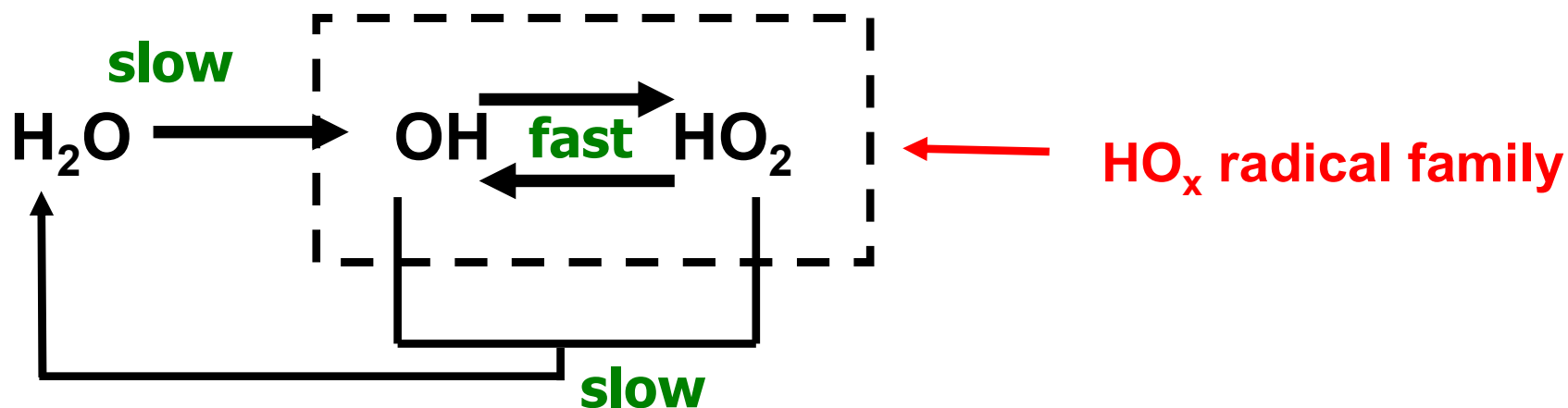
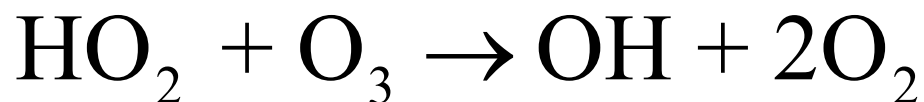
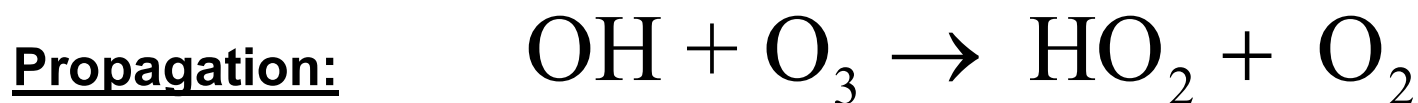
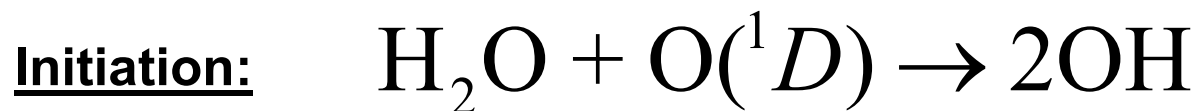
WATER VAPOR IN STRATOSPHERE

Aura MLS Equivalent Latitude/ θ Global Section -- 6 Nov 2008 (2008d311)

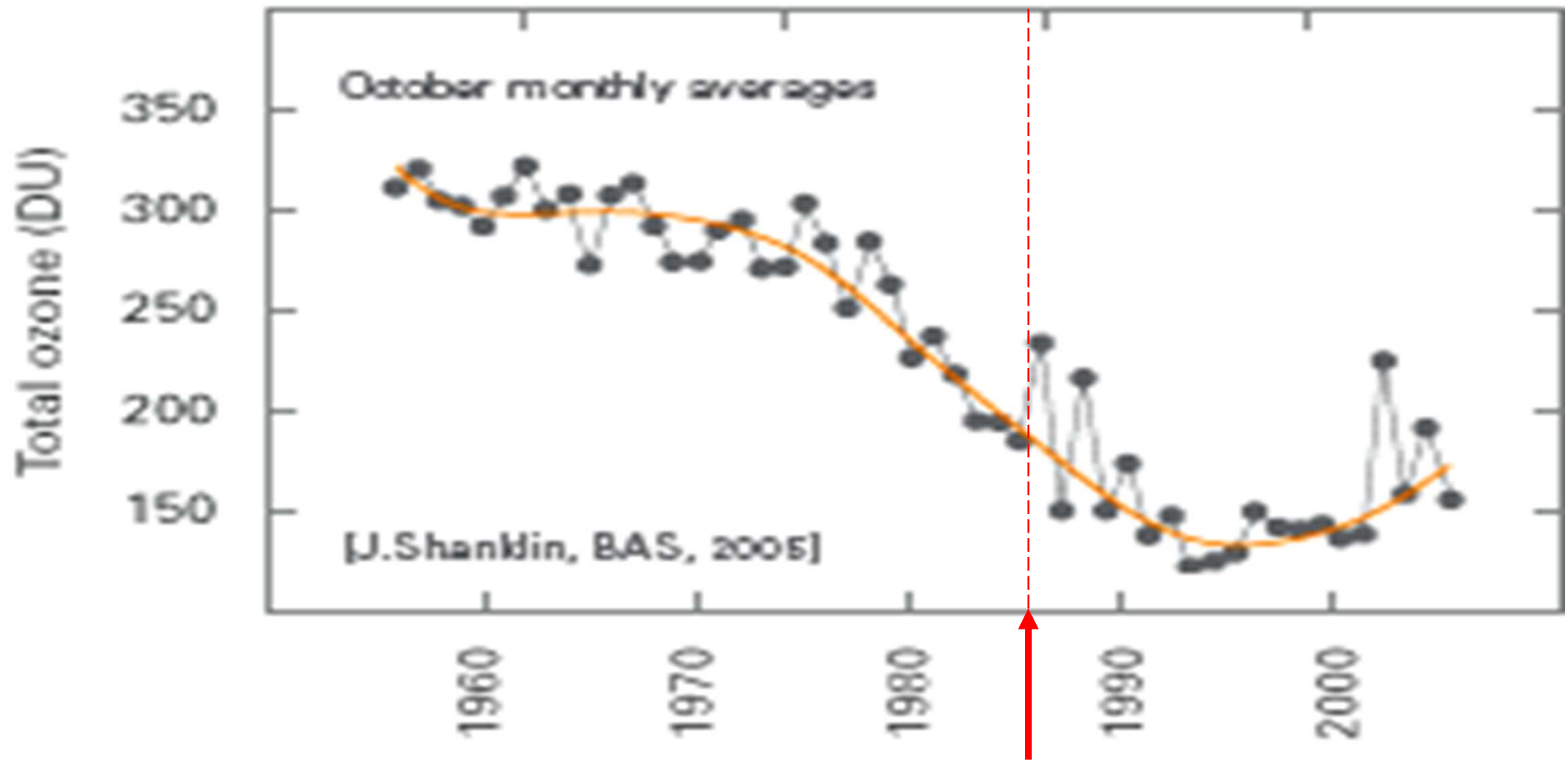


Source: transport from troposphere, oxidation of methane (CH₄)

Ozone loss catalyzed by hydrogen oxide ($\text{HO}_x \equiv \text{H} + \text{OH} + \text{HO}_2$) radicals



OZONE TREND AT HALLEY BAY, ANTARCTICA (OCTOBER)

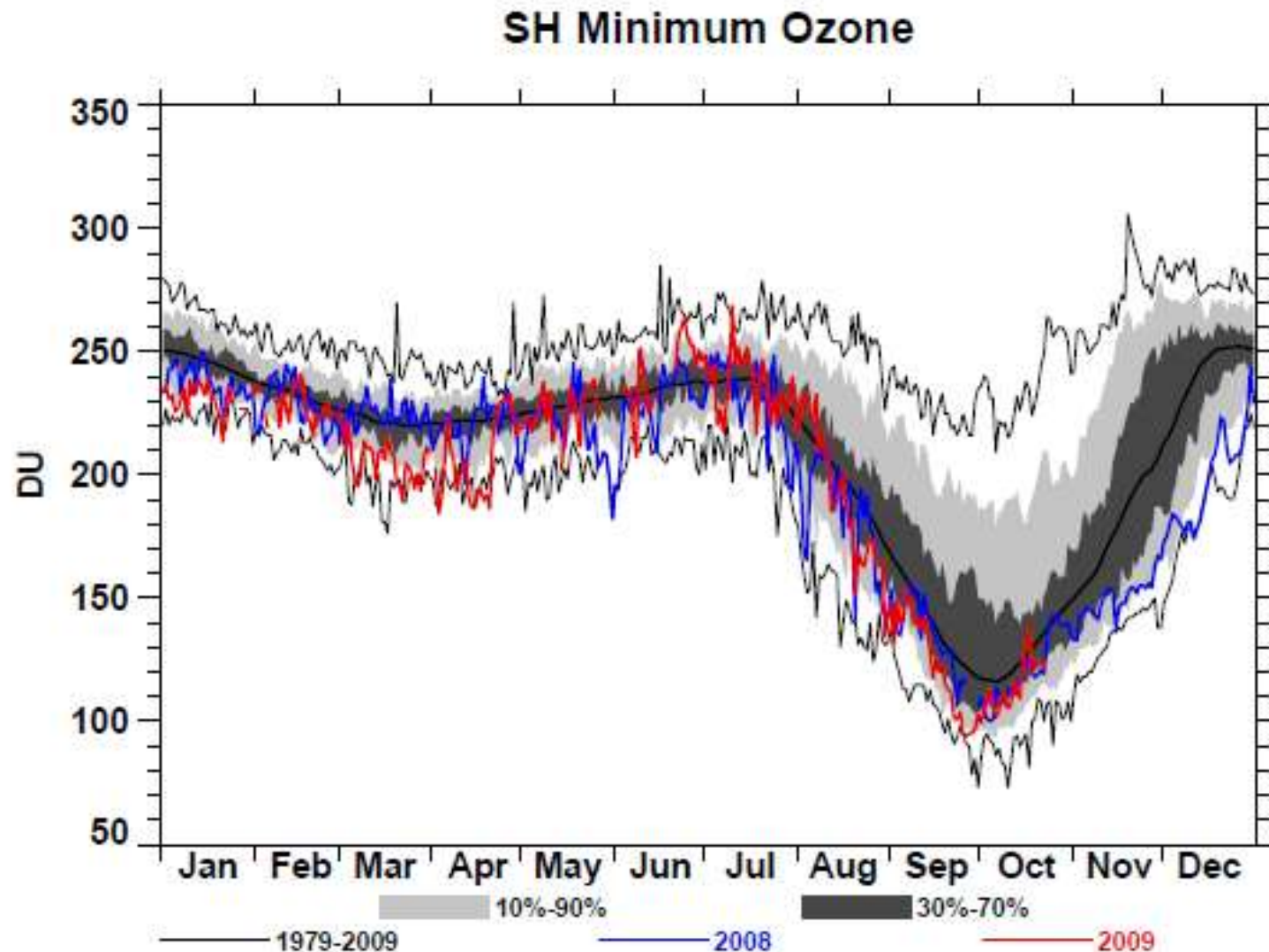


Recognized late because the data processing threw out outliers

Farman et al. 1985 paper published in *Nature*

1 Dobson Unit (DU) = 0.01 mm O₃ STP = 2.69×10^{16} molecules cm⁻²

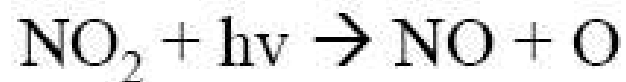
THE OZONE HOLE IS A SPRINGTIME PHENOMENON



SLIDE: DANIEL JACOB

CATALYSIS BY NO_x : WORK DONE IN 1960s-1970s

NOBEL PRIZE 1995, PAUL CRUTZEN



THE ABOVE IS A
NULL CYCLE FOR O₃

BUT SOME NO₂ CAN REACT WITH O SINGLET;



TERMINATION STEP



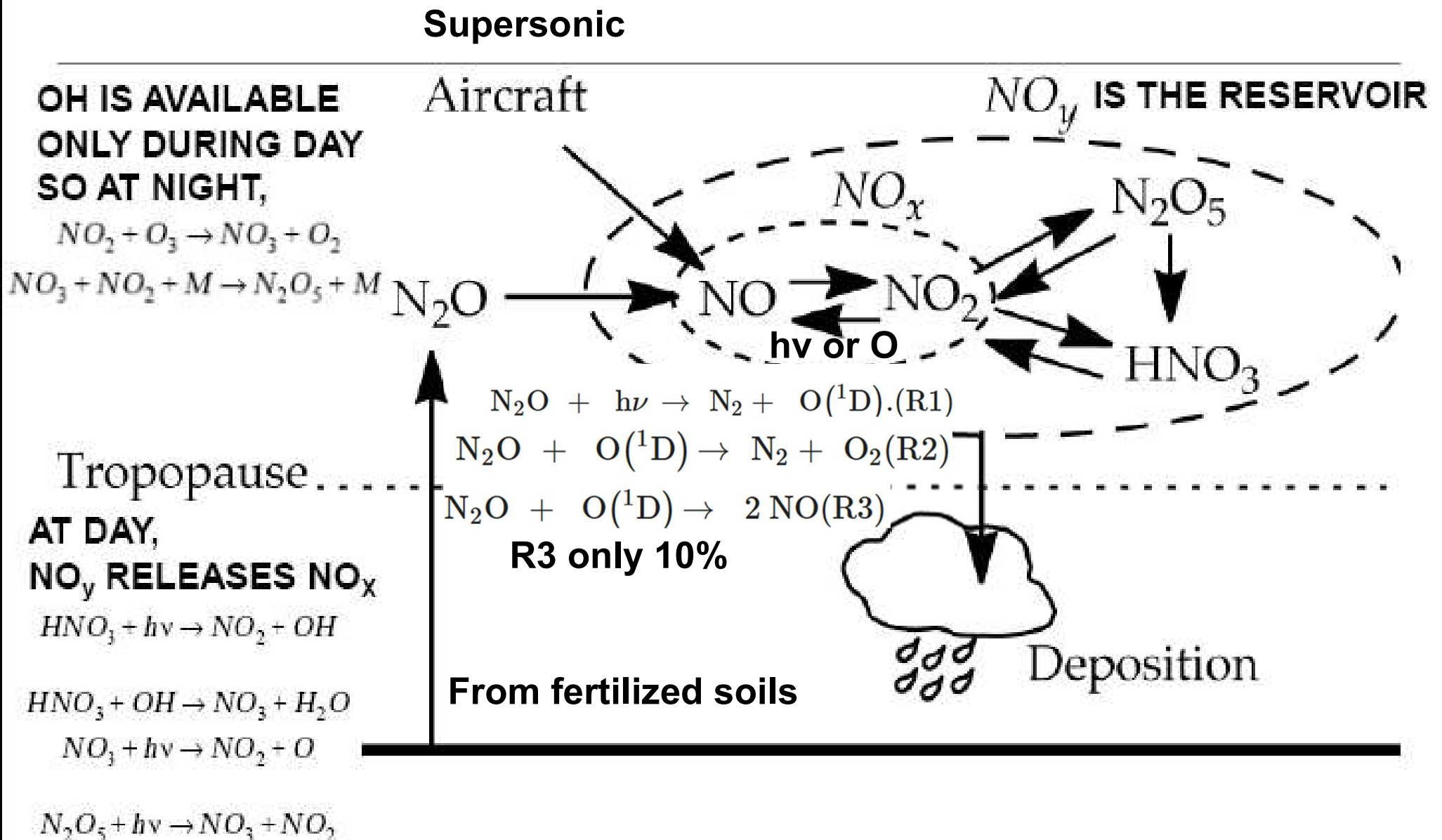
Normal way of
ozone formation in
the troposphere

EACH CYCLE CONSUMES
TWO O_x MOLECULES
RATE LIMITING STEP IS
THIS REACTION BECAUSE
NO₂ CAN EITHER
PHOTOLYZE OR REACT WITH
O

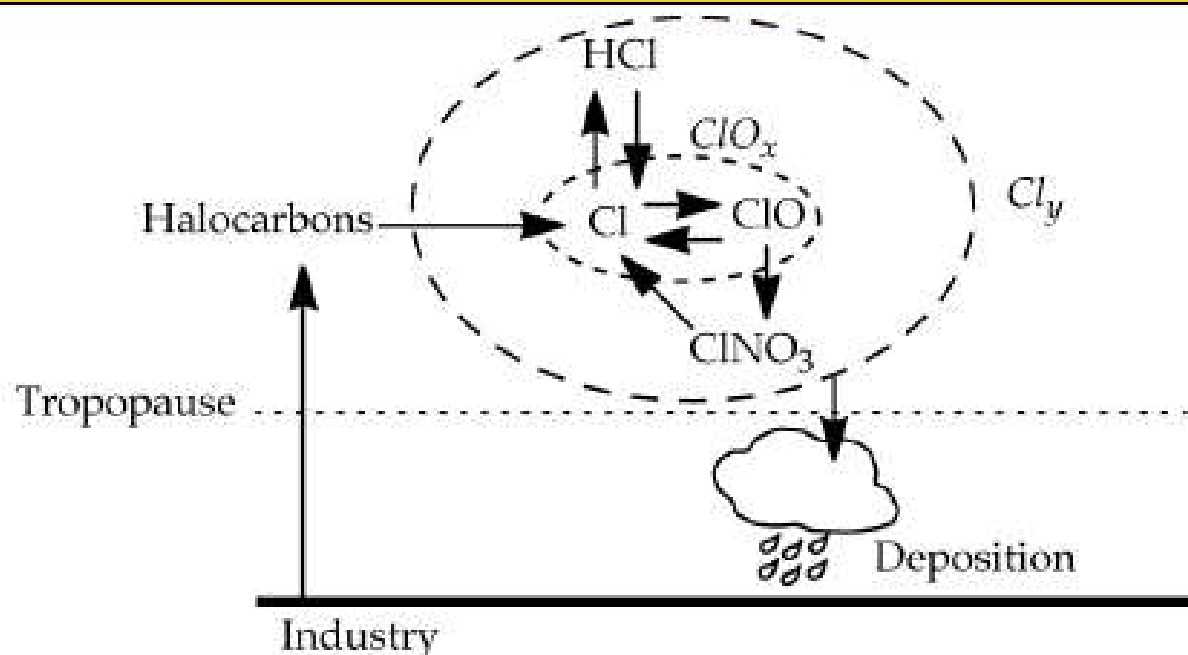
✓ Together with
HO_x catalysis
closed the
budget of
Stratospheric
ozone

✓ Along with
Johnston's work
led to scrapping of
supersonic fleet in
USA in early 1970s
though Concorde
in Europe still
continued for some
time

ATMOSPHERIC CYCLING OF NO_x AND NO_y



CHLORINE RADICALS (ClO_x & Cl_y)



IN 1974, MARIO MOLINA
& SHERWOOD ROWLAND
POINTED OUT THE
POTENTIAL FOR OZONE
LOSS DUE TO CFCs;
CO-WINNERS OF NOBEL
PRIZE WITH PAUL
CRUTZEN

FOR EXAMPLE, CF_2Cl_2 (TRADE NAME CFC-12); Very long lived; transported to Stratosphere and undergoes photolysis there:



THEN SIMILAR TO NO_x IT CATALYTICALLY DESTROYS OZONE:



THE MONTREAL PROTOCOL: 1987

MOLINA & ROWLAND WARNED THAT ClO_x CATALYZED O_3 LOSS WOULD DAMAGE THE OZONE LAYER IF EMISSIONS OF CFCs DID NOT STOP. IN 1985 OBSERVATIONS OF OZONE HOLE IN ANTARCTICA WERE REPORTED IN NATURE BY FARMAN ET AL. A SERIES OF INTERNATIONAL AGREEMENTS TO BAN CFCs FOLLOWED WHICH FINALLY LED TO THE MONTREAL PROTOCOL IN 1987

The Montreal Protocol is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances believed to be responsible for ozone depletion. The treaty was opened for signature on September 16, 1987, & entered into force on January 1, 1989, followed by a first meeting in Helsinki, May 1989. Since then, it has undergone seven revisions, in 1990 (London), 1991 Nairobi 1992 (Copenhagen), 1993 (Bangkok), 1995 (Vienna), 1997 (Montreal), and 1999 (Beijing). It is believed that if the international agreement is adhered to, the ozone layer is expected to recover by 2050! It has been ratified by 196 states

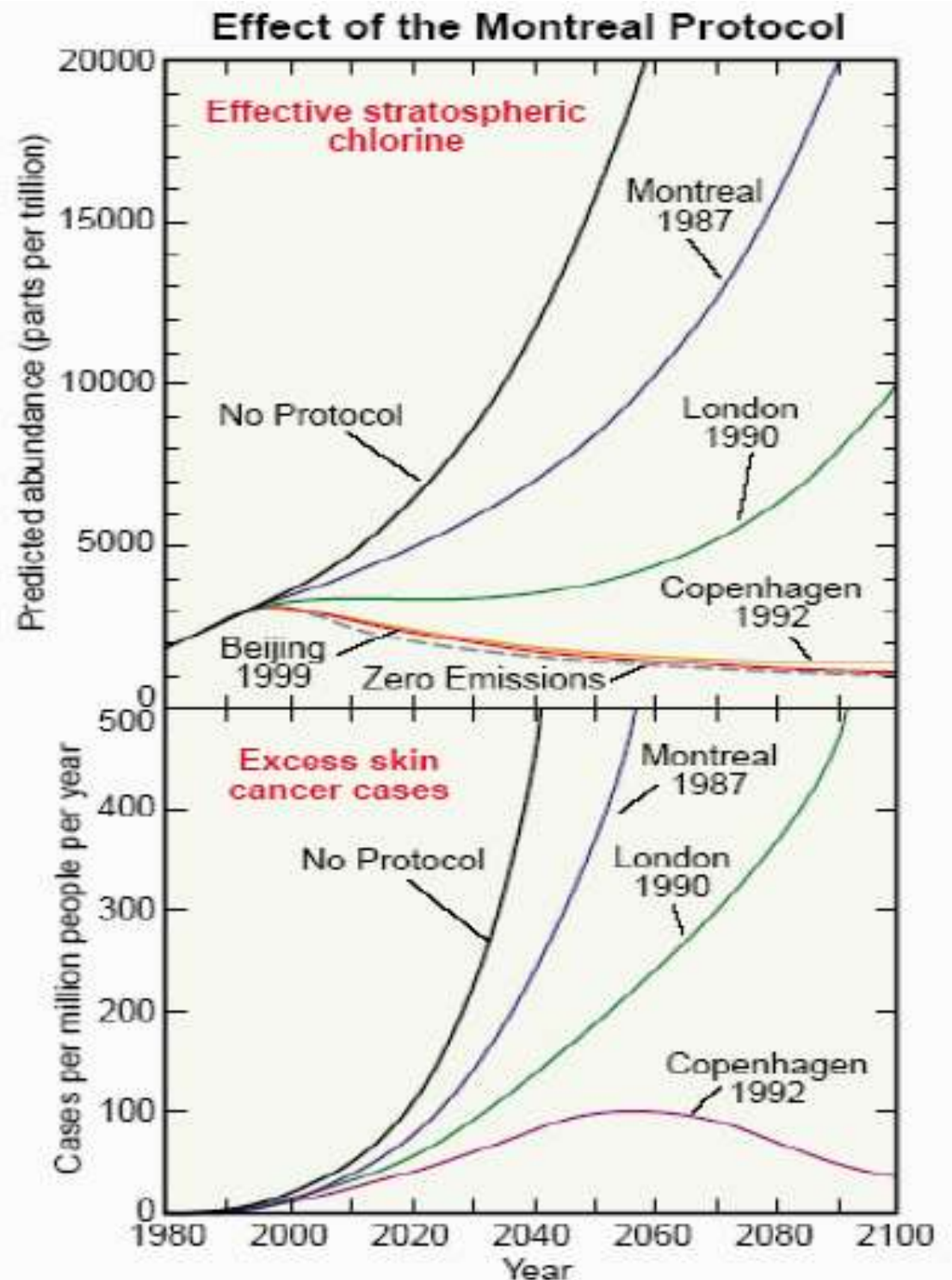
CFCl_3 (CFC-11), CF_2Cl_2 (CFC-12), $\text{C}_2\text{F}_3\text{Cl}_3$ (CFC-113), $\text{C}_2\text{F}_4\text{Cl}_2$ (CFC-114)
and $\text{C}_2\text{F}_5\text{Cl}$ (CFC-115)

WAS IT EASY TO CONVINCE PEOPLE USING GOOD SCIENCE ?

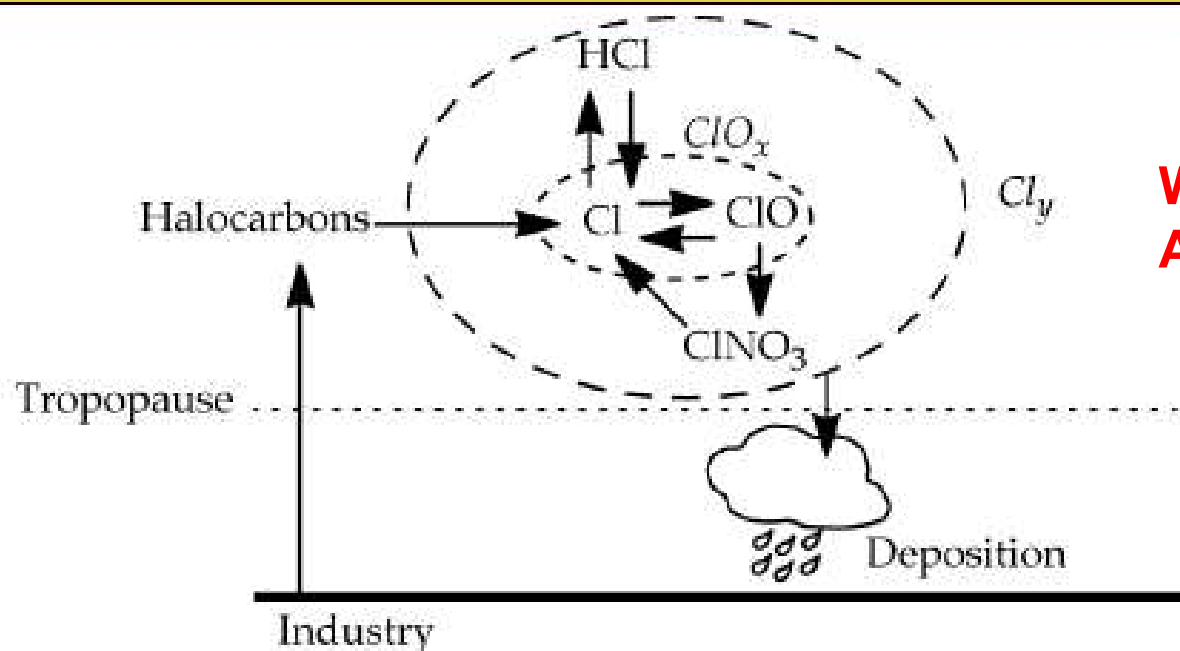
“The Rowland-Molina hypothesis was strongly disputed by representatives of the aerosol and halocarbon industries. The chair of the board of DuPont was quoted as saying that ozone depletion theory is "a science fiction tale...a load of rubbish...utter nonsense “

- After publishing their pivotal paper in **June 1974**, Rowland and Molina testified at a hearing before the U.S. House of Representatives in **December 1974**. As a result significant funding was made available to study various aspects of the problem and to confirm the initial findings. In 1976, the U.S. National Academy of Sciences (NAS) released a report that confirmed the scientific credibility of the ozone depletion hypothesis
- Then, **in 1985**, British Antarctic Survey scientists Farman, Gardiner and Shanklin shocked the scientific community when they published results of a study showing an ozone "hole" in the journal Nature — showing a decline in polar ozone far larger than anyone had anticipated.

SKIN CANCER EPIDEMIOLOGY PREDICTIONS



CHLORINE RADICALS (ClO_x & Cl_y)



Why was the ozone hole large?
A sink that isn't a sink!

TERMINATION STEP: CONVERSION OF ClO_x TO NON-RADICAL CHLORINE RESERVOIRS, HCl & ClONO_2

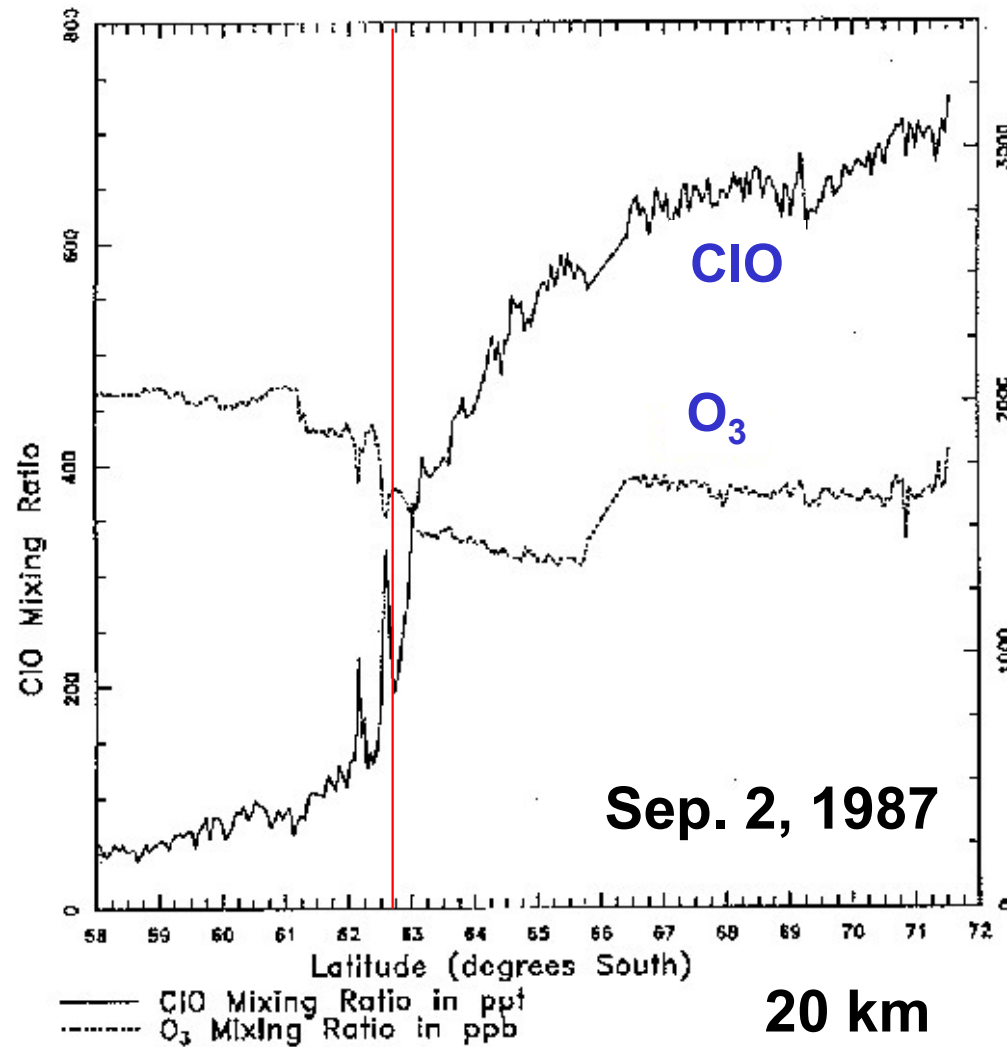


EVENTUALLY THOUGH,

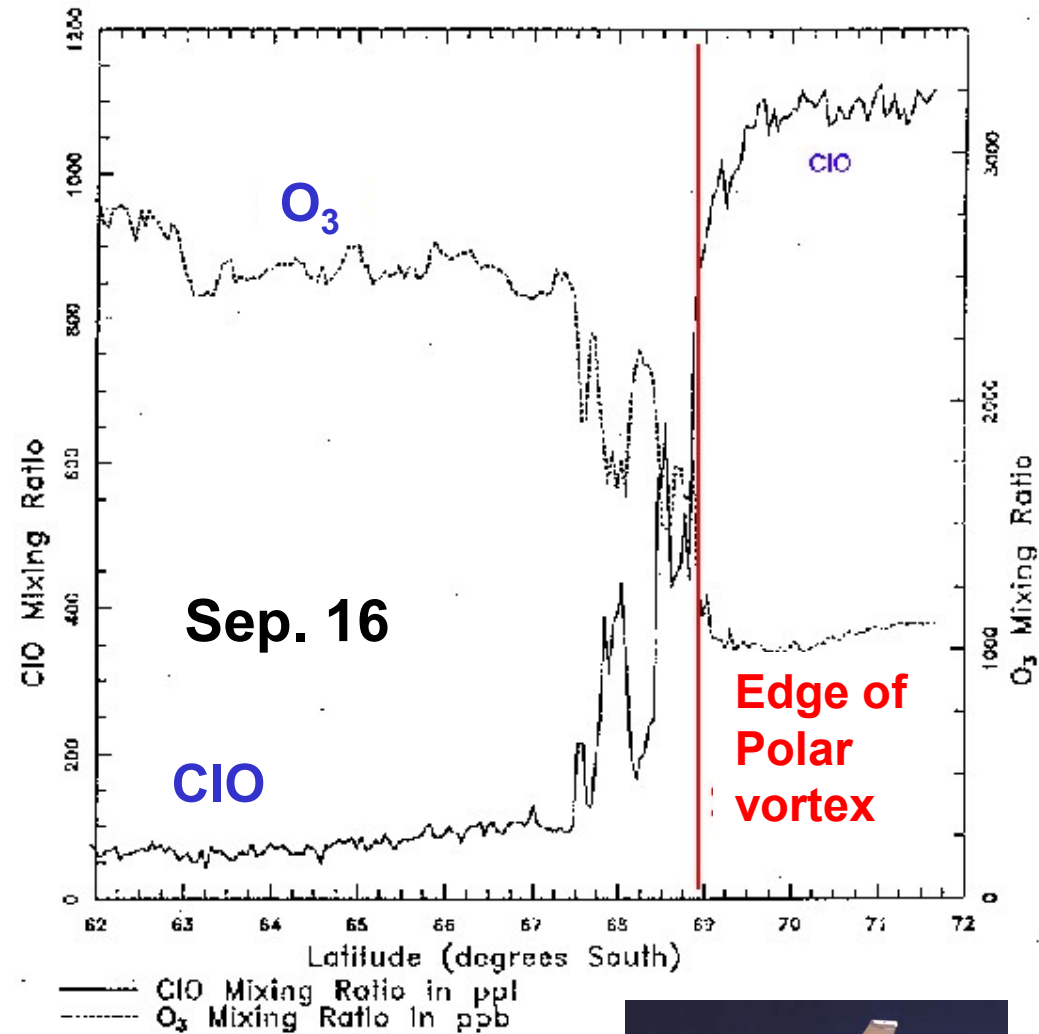


ASSOCIATION OF ANTARCTIC OZONE HOLE WITH HIGH LEVELS OF ClO

Sept. 1987 ER-2 aircraft measurements at 20 km altitude south of Punta Arenas



20 km
altitude



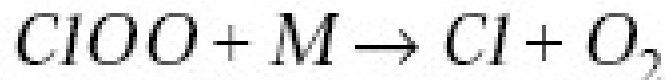
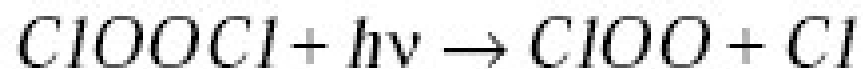
Measurements by Jim Anderson's group (Harvard)



A MYSTERY IN THE CHEMISTRY

Studies found depletion of Ozone to be associated with Exceptionally high ClO, a result confirmed by satellite data

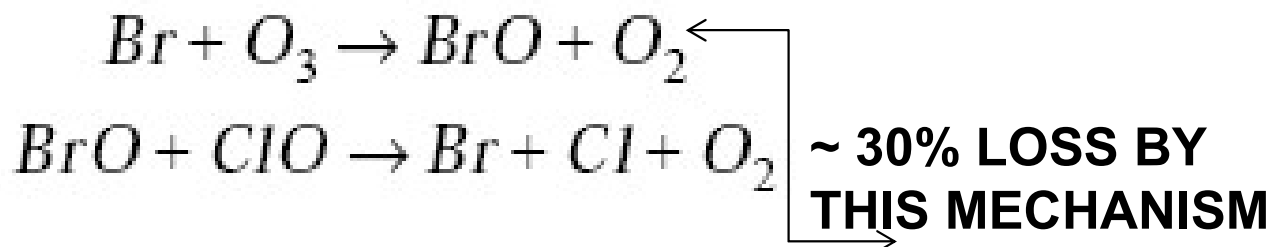
Parallel laboratory experiments revealed that at such high ClO levels, a new catalytic cycle involving self reaction of ClO could account for most observed Ozone depletion



**KEY DISCOVERY HERE
WAS THAT PHOTOLYSIS OF ClOOC1
OCCURS AT O-Cl BOND RATHER
THAN O-O BOND WHICH IS WEAKER
NOTE IT DOES NOT DEPEND ON
[O]**



AND IF Br IS PRESENT,

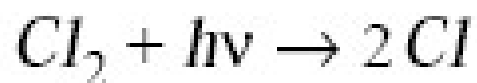
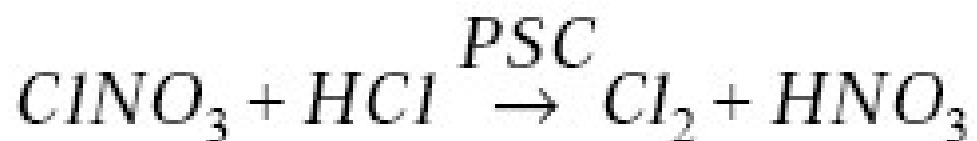


WHY THE HIGH CIO IN THE ANTARCTIC VORTEX?

Chlorine radicals from reactions of Reservoir species in (PSCs)

RESEARCH IN 1990s SHOWED THE ROLE OF STRATOSPHERIC AEROSOLS AT LOW TEMPERATURE

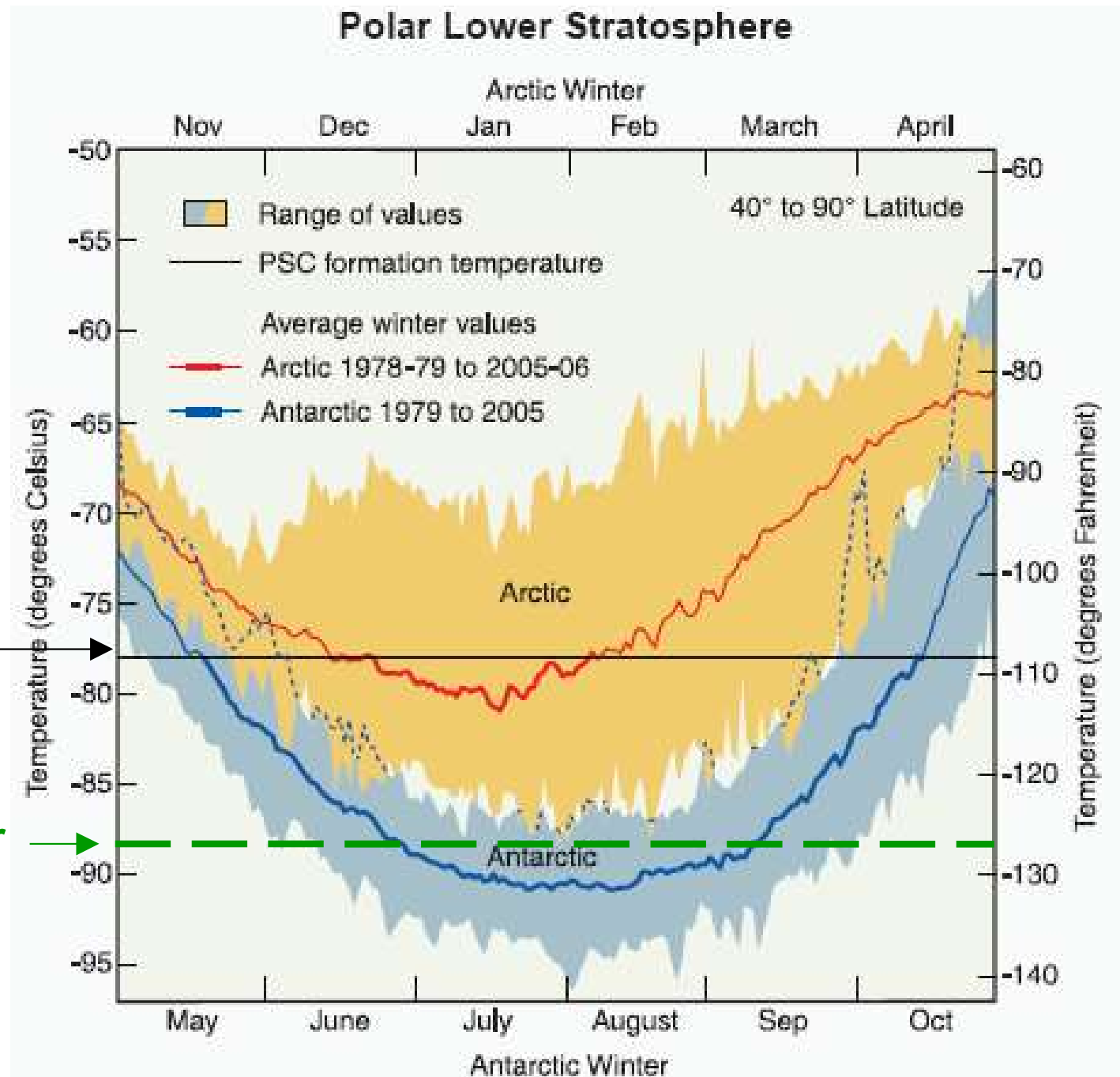
TEMPERATURES IN WINTERTIME ANTARCTIC STRATOSPHERE ARE LOW ENOUGH TO CAUSE FORMATION OF PSCs WHICH PROVIDE **SURFACES** FOR CONVERSION OF ClO_x RESERVOIRS HCl & ClNO_3 TO Cl_2 WHICH THEN RAPIDLY PHOTOLYZES TO PRODUCE Cl



WHY ARE PSCs FORMED IN THE ANTARCTIC?

- **Stratosphere is extremely dry and so condensation of water vapour to form ice clouds requires very low temperatures**
 - **3-5 ppmV corresponds to water vapour pressure of $3\text{-}5 \times 10^{-4}$ hPa at which the frost point is 185-190 K & such low temperatures rarely occur (only during antarctic winter)**
 - **When temperatures do fall below 190 K, PSC's are observed which form already at 197 K, and this higher threshold for PSC formation is responsible for the large extent of the O₃ hole**
 - **Stratosphere also contains enough HNO₃ so that solid HNO₃-H₂O phases may form, namely, NAT, NAD & NAM**
- LACK OF TOPOGRAPHY & DIFF. LAND SEA HEATING
LEADS TO VERY LITTLE MERIDIONAL TRANSPORT & EFF.
ISOLATION OF THE ANATRCTIC WHICH IS STRONGEST IN
WINTER**

PSC FORMATION AT COLD TEMPERATURES

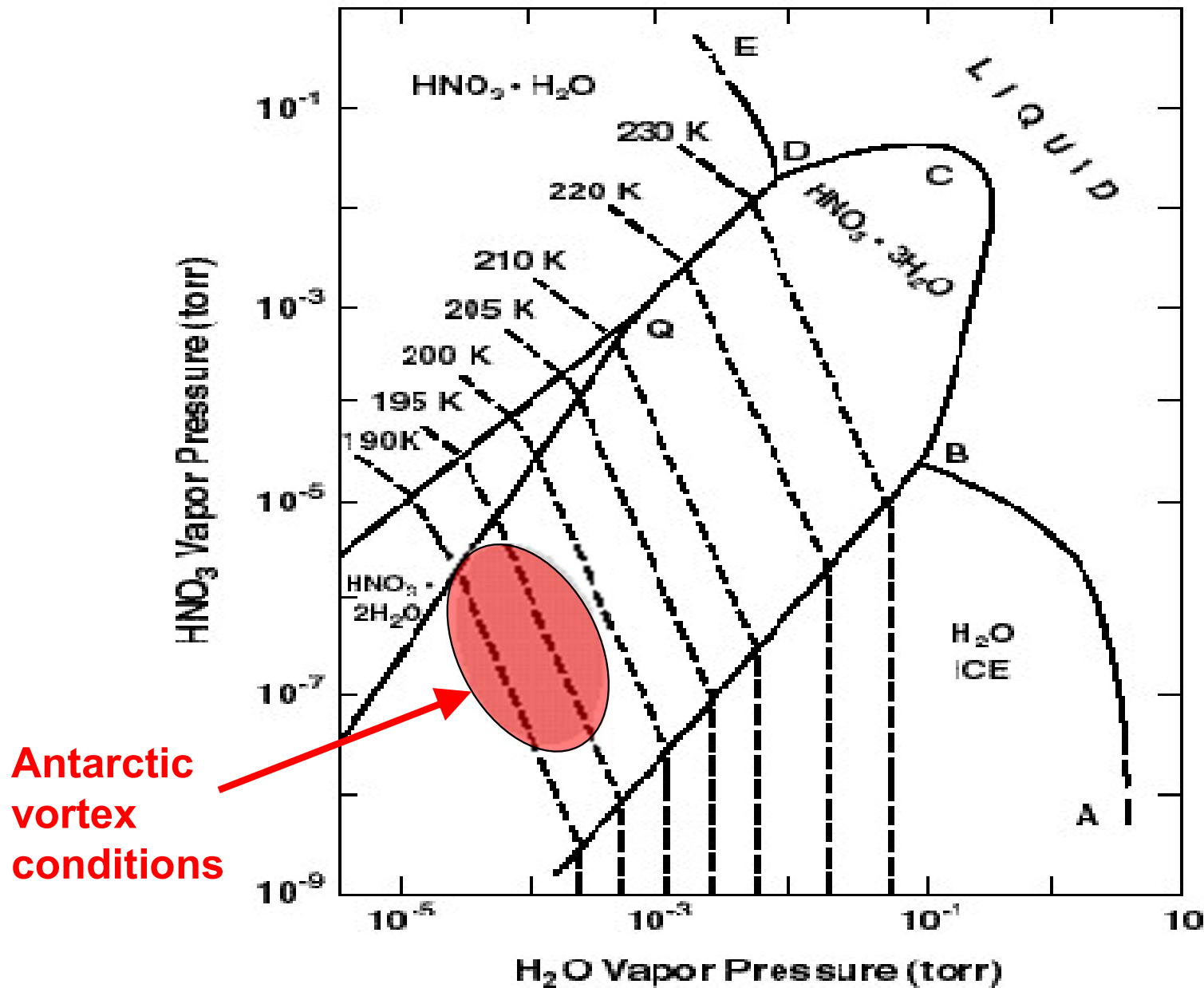


PSC formation

Frost point of water

HOW DO PSCs START FORMING AT 195K?

HNO₃-H₂O PHASE DIAGRAM



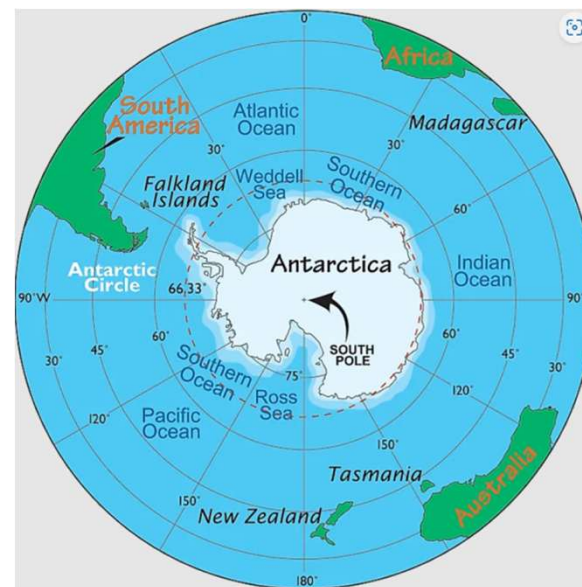
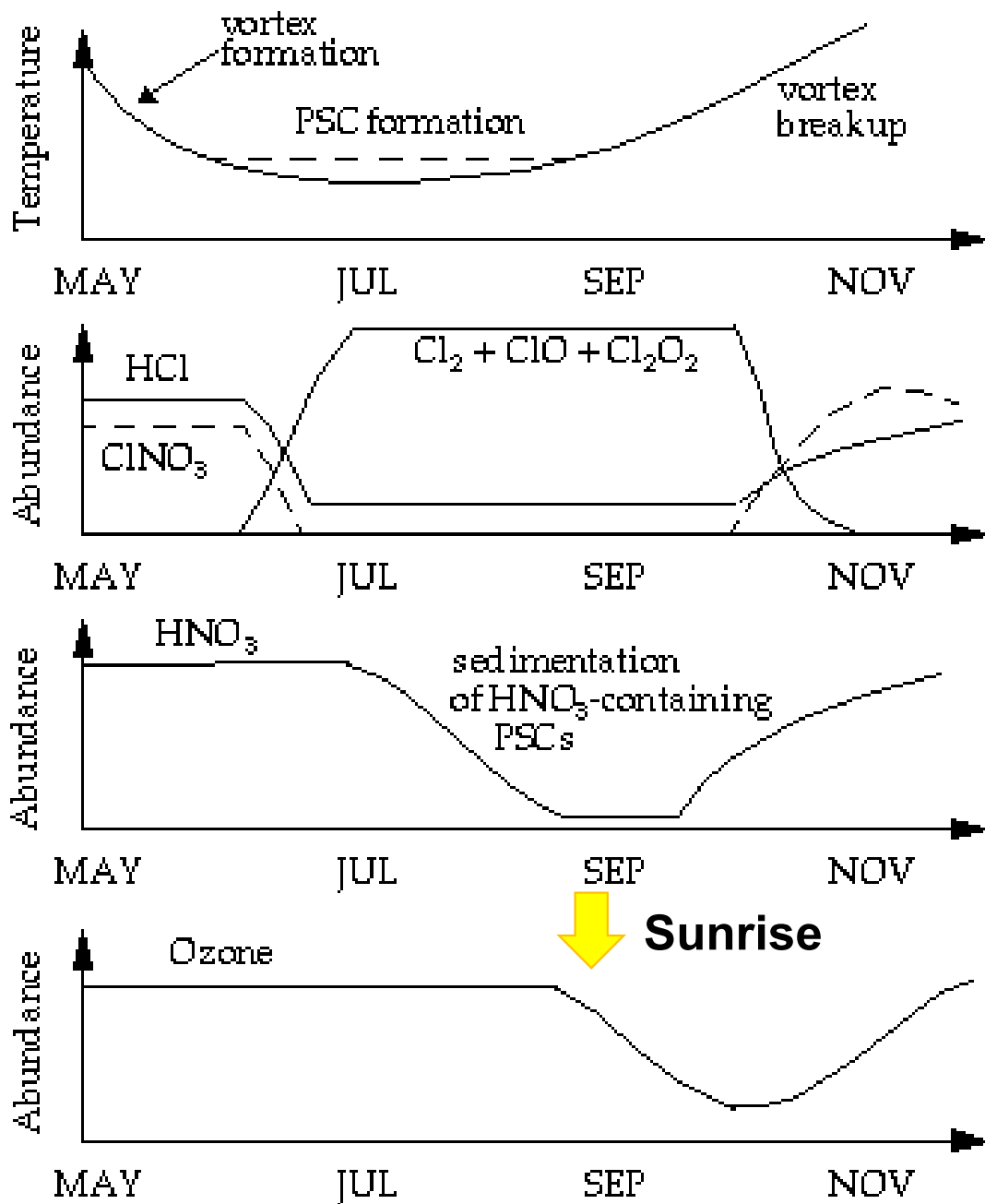
6 phases
H₂O ice
Liq. HNO₃-H₂O
Gas HNO₃-H₂O
NAT
NAD
NAM

Number of n
Independent
variables
defining the
T.D stable
system is

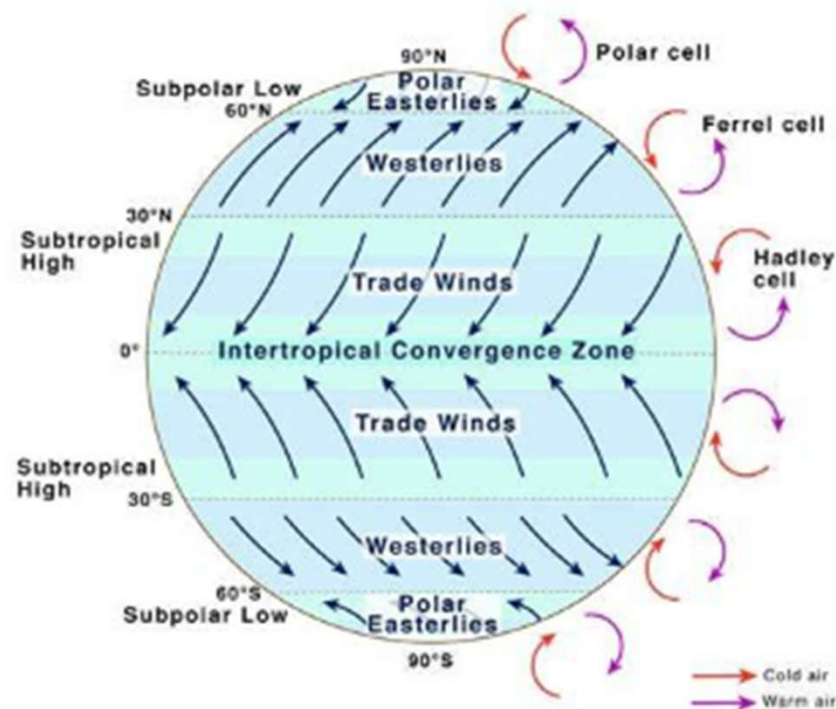
$n = c - p + 2$
Eqb of PSC with
Gas phase
 $= 2 + 2 - 2$
 $= 2$, so given
HNO₃ (g), H₂O(g)
we can infer fav.

PSCs are not water but nitric acid trihydrate (NAT) clouds phase

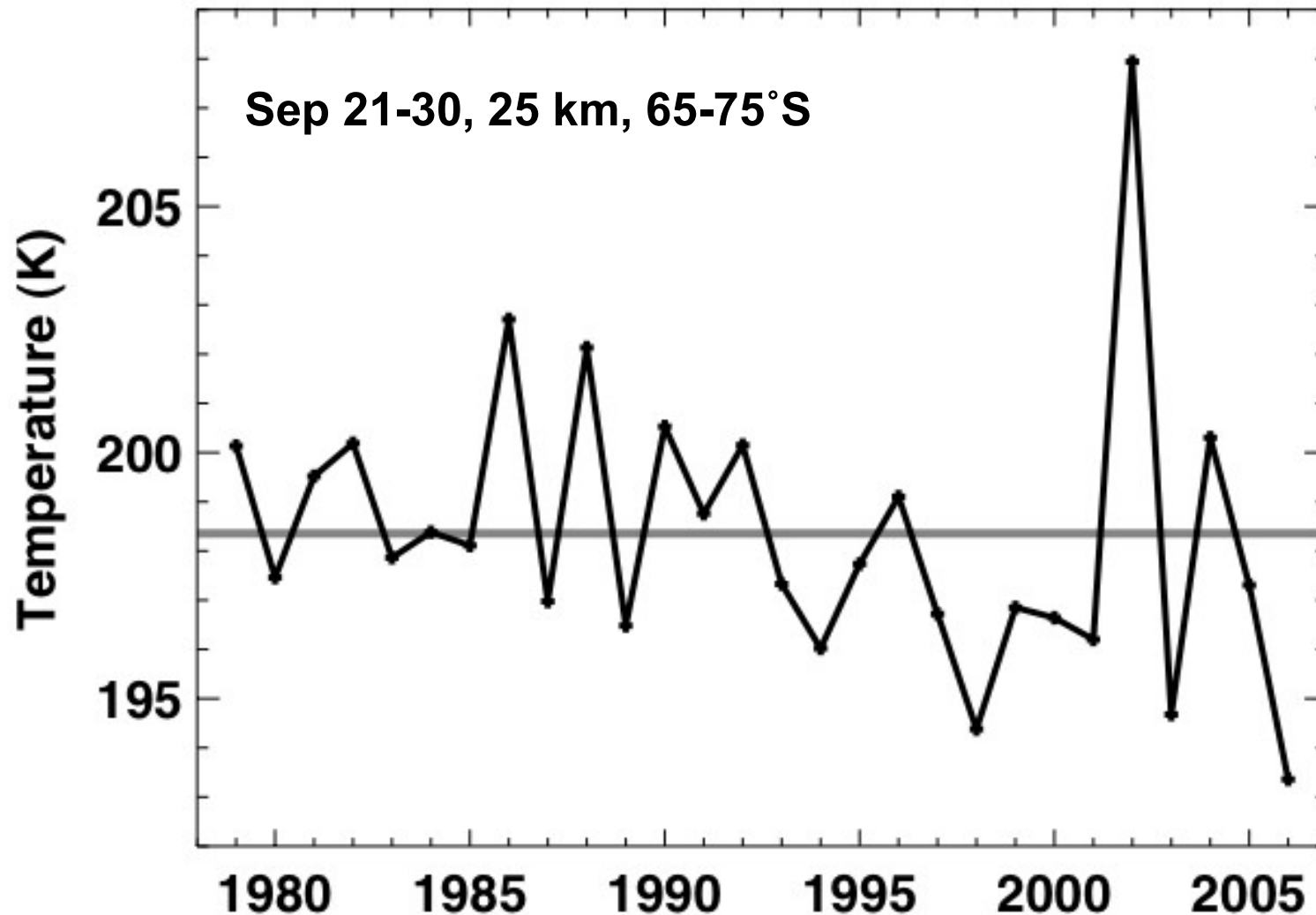
CHRONOLOGY OF ANTARCTIC OZONE HOLE



Global Wind Patterns



LONG-TERM COOLING OF THE STRATOSPHERE

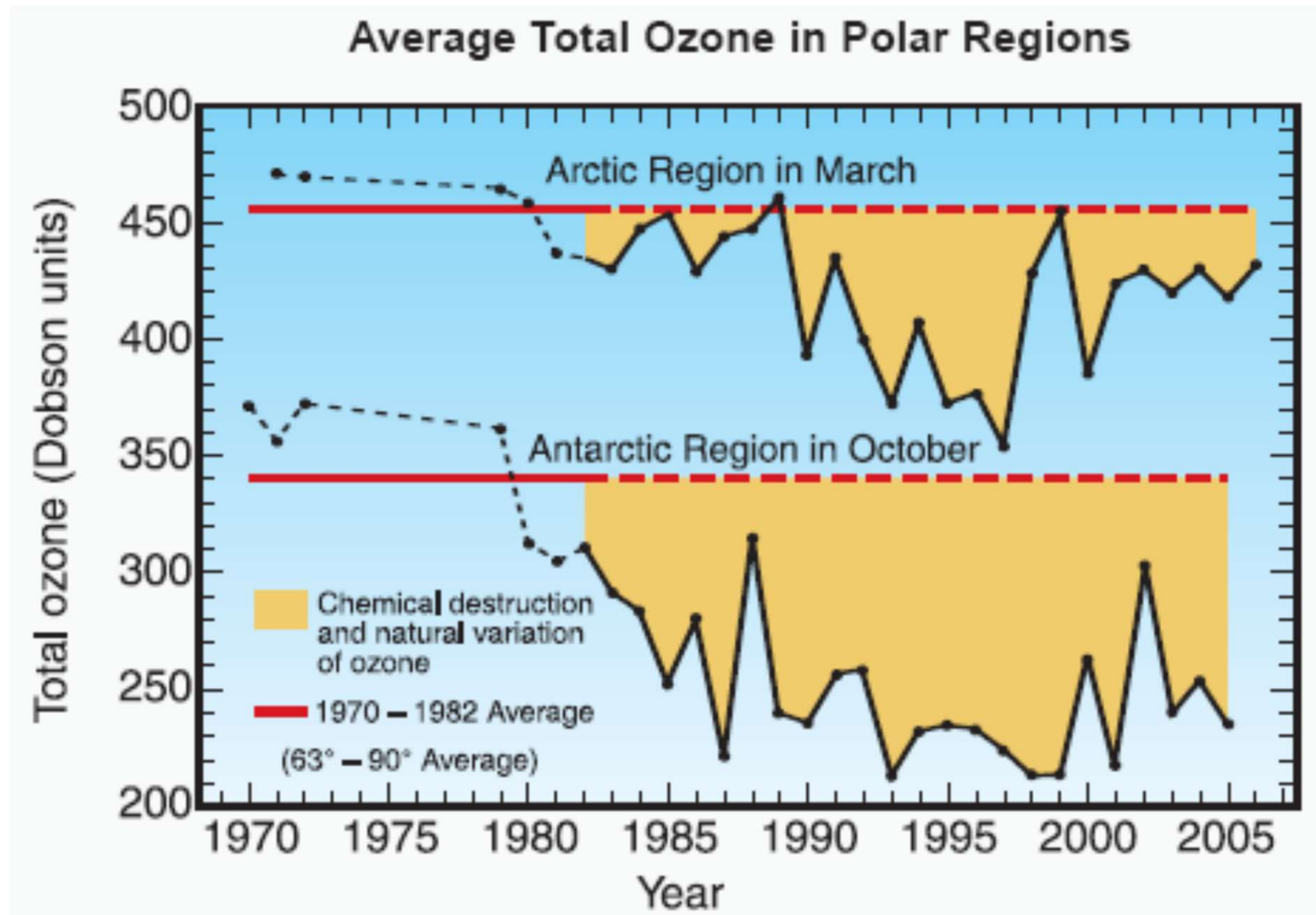


Increasing CO₂ is expected to cool the stratosphere

TRENDS IN POLAR OZONE

Could greenhouse-induced cooling of stratosphere produce an Arctic ozone hole over the next decade?

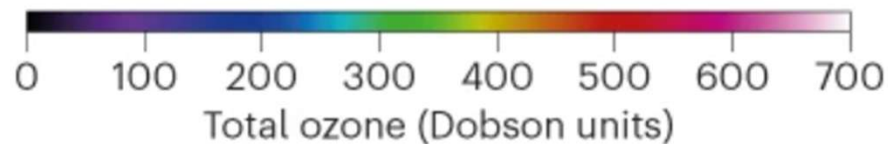
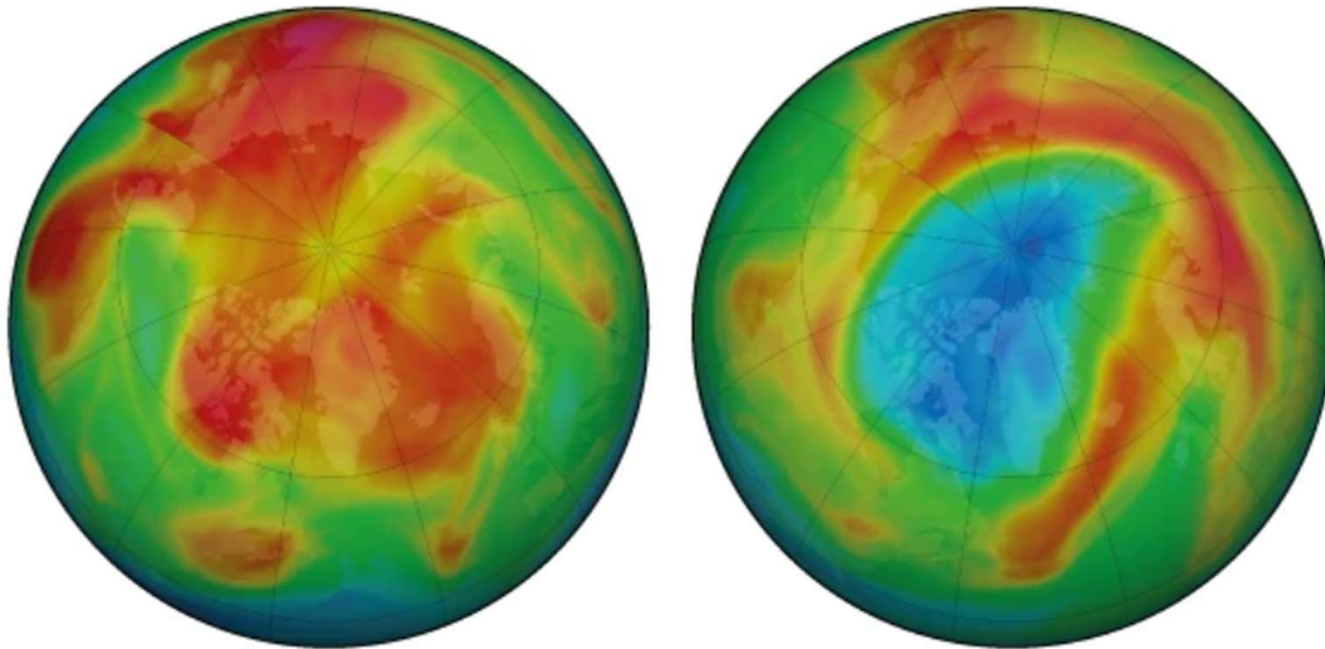
Race between chlorine decrease and climate change



Large holes were seen in 1997, 2011, 2020

23 March 2019

23 March 2020



©nature

Manney, G. L. *et al. Nature* **478**, 469-475 (2011).
Nature **580**, 18-19 (2020)

PRACTICE QUESTIONS

Discuss during lab if you have doubts

Numerical

1. The bond energy of the of the O₂ double bond is kJ/mol what is the minimum wavelength required to photolyse the bond? ($N_A=6.022 \times 10^{23}$, $h=6.62607015 \times 10^{-34}$ [J·Hz⁻¹] or [J·s], $c= 299792458$ m s⁻¹) You may want to revise Lecture 11 to answer this type of questions!

Short/long answer style

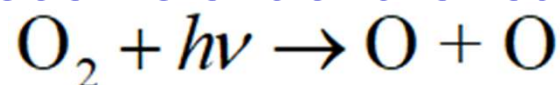
1. Write the four reactions that make up the Chapman cycle
2. Write how NO is formed in the stratosphere and show how many molecules of ozone one molecule of NO destroys in the stratosphere. How would a fleet of supersonic aircrafts affect the ozone layer?
3. What does CFC stand for? Why do CFCs not react with ozone in the troposphere? Why was their manufacturing and sale banned via the Montreal Protocol?
4. Write the reactions that show how chlorine catalytically destroys ozone. Which role do polar stratospheric clouds (PSCs) play in the formation of the ozone hole. Why is ClOOCl formed in PSCs but not in the upper stratosphere?

5. Explain why the rate coefficient of the reaction:



is pressure dependent. Is the reaction slower or faster in the lower stratosphere.

6. Explain why the rate coefficient of the reaction:



is altitude dependent. Is the reaction slower or faster in the lower stratosphere.